Data Article

Title: Hierarchical forecasting of COVID-19 cases in Africa using machine learning models

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Abstract
The COVID-19 pandemic severely challenged public health systems in Africa due to data scarcity, weak infrastructure, and regional disparities. To address the need for accurate, coherent COVID-19 forecasts at continental, regional, and national levels, this study developed a hierarchical time-series forecasting model using a bottom-up reconciliation approach enhanced by machine learning. Specifically, extreme gradient boosting (XGBoost) and random forest models were combined via a weighted ensemble, with forecasts generated at the national level and aggregated upward. Compared with ARIMA and exponential smoothing, XGBoost alone performed best. However, the ensemble method that assigned greater weight to XGBoost achieved the lowest mean absolute error, root mean square error and mean absolute scale error. Notably, Southern Africa reported the highest case counts despite low population density, revealing underlying health and socioeconomic vulnerabilities. The bottom-up machine learning approach proved effective for data-limited settings. The data are stored in an Excel file.

Specifications Table

Subject area	*Hierarchical Time Series Forecasting*
More specific subject area	*Infectious Diseases Modelling and Forecasting*
Type of data	*Excel file*
How data was acquired	*Provided from the internet*
Data format	*Filtered and analysed.*
Experimental factors	*N/A*
Experimental features	*N/A*
Data source location	*https://www.worldometers.info/coronavirus/*
Data accessibility	*The data is hosted on GitHub https://github.com/csigauke/Hierarchical-forecasting-of-COVID-19-cases-in-Africa-using-Machine-Learning-models*
Related research article	Hierarchical forecasting of COVID-19 cases in Africa using machine learning models

Value of the Data

The value of this dataset lies in its comprehensive, long-term coverage and regional specificity for the African continent. Spanning three full years (2020-02-16 to 2023-02-18), the data captures the evolution of the COVID-19 pandemic from its early stages through multiple variant waves (including Delta and Omicron) and into the endemic phase, enabling robust trend analysis and modelling. Its critical value is the segmentation into Africa's five distinct sub-regions, Northern, Eastern, Southern, Central, and Western, enabling direct comparative epidemiological studies to understand why different parts of the continent experienced varying transmission dynamics. Furthermore, including population sizes for each region transforms raw case counts into standardised incidence rates (cases per 100,000 people), facilitating fair, demographically adjusted comparisons of disease burden across areas with vastly different population sizes. This combination of a complete, three-year time series, regional disaggregation, and population denominators makes the data exceptionally valuable to public health officials and researchers. It can be used to guide evidence-based resource allocation for future health emergencies, train forecasting models, and inform targeted intervention strategies across Africa's diverse epidemiological landscapes.

Data

The data used in this study are daily confirmed COVID-19 cases. The study covers the period from February 16, 2020, to February 18, 2023, and focuses on five regions of Africa: Northern (NA), Eastern (EA), Southern (SA), Central (CA), and Western (WA). The data is structured hierarchically, with country-level information aggregated to regional levels and to a total for Africa. The population data used in the study includes the following figures: NA with 272,131,339; EA with 500,703,846; SA with 73,138,701; CA with 212,915,636; and WA with 456,251,329.

Experimental Design, Materials, and Methods

For a detailed discussion of the methods used, we refer to the published article, which can be accessed at https://doi.org/10.3389/fepid.2026.1696282

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References

Worldometer https://www.worldometers.info/coronavirus/ (Accessed on 13 January 2025).